

Silicon Photonic Mach-Zehnder Interferometer Design Proposal

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I. INTRODUCTION

This proposal presents the design and simulation of silicon photonic Mach-Zehnder interferometers (MZIs) for extracting the group index of a strip waveguide. An MZI divides the input optical field into two paths and recombines them after introducing a controlled path-length difference. The resulting wavelength-dependent interference provides a direct way to determine the free spectral range (FSR), which can be used to calculate the waveguide group index.

The design uses a 500 nm wide and 220 nm high silicon strip waveguide with quasi-TE polarization, Y-branch splitters, and TE grating couplers. Six MZI circuits are included so that different path-length imbalances can be compared, while MZI6 repeats MZI1 to provide a nominally identical device for assessing fabrication variation.

II. WAVEGUIDE

The waveguide is a rectangular silicon channel surrounded by silicon dioxide. Its width is 500 nm and its height is 220 nm. The fundamental quasi-TE mode was calculated using the finite-difference eigenmode (FDE) solver in Lumerical MODE at the design wavelength of 1550 nm. The first mode has $n_{\text{eff}} = 2.445503$ and a TE polarization fraction of approximately 98%.

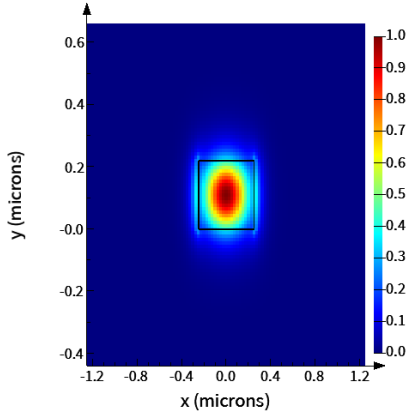


Fig. 1. Fundamental quasi-TE mode profile of the 500 nm × 220 nm silicon waveguide at 1550 nm.

The effective index was tracked over 1.50–1.60 μm. The simulated n_{eff} decreases with increasing wavelength, showing the expected waveguide and material dispersion.

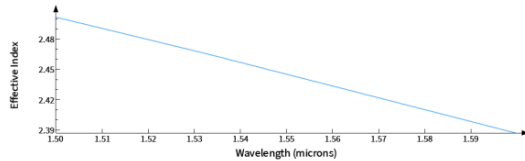


Fig. 2. Effective index of the fundamental quasi-TE mode versus wavelength.

The group index is obtained from the effective-index dispersion as

$$n_g = n_{\text{eff}} - \lambda(dn_{\text{eff}}/d\lambda).$$

The simulated group index changes smoothly over the same wavelength range and is approximately 4.194 at 1550 nm.

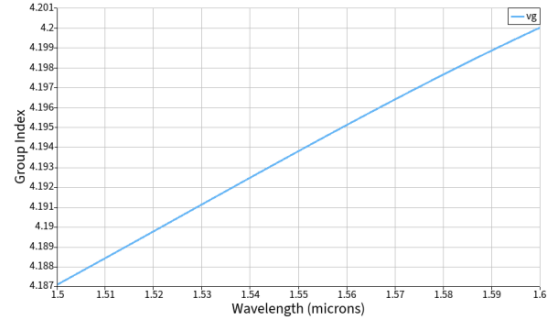


Fig. 3. Group index of the fundamental quasi-TE mode versus wavelength.

A compact polynomial was fitted to the effective-index data so that the waveguide dispersion can be represented without rerunning the eigenmode calculation at every wavelength:

$$n_{\text{eff}}(\lambda) = 2.44368868 - 1.12905725(\lambda - 1.55) - 0.04221652(\lambda - 1.55)^2.$$

Here, λ is in micrometers and the model is centered at $\lambda_0 = 1.55 \mu\text{m}$.

III. MACH-ZEHNDER INTERFEROMETER DESIGN

For an imbalanced MZI, the path-length difference is defined as $\Delta L = L_2 - L_1$. The phase difference between the two arms is

$$\Delta\phi = \beta\Delta L = (2\pi n_{\text{eff}}/\lambda)\Delta L.$$

For ideal 50:50 splitting and recombination, the normalized transfer function is

$$T(\lambda) = I_{\text{out}}/I_{\text{in}} = 1/2[1 + \cos(\beta\Delta L)].$$

The FSR is determined by the group-delay difference between the two arms:

$$\text{FSR} = \lambda^2/(n_g \Delta L).$$

Therefore, the group index can be extracted directly from a measured or simulated FSR:

$$n_g = \lambda_0^2/[\Delta L(\text{FSR})].$$

TABLE I
MZI PATH-LENGTH VARIATIONS AND EXPECTED FSR AT 1550 nm

MZI	L_1 (μm)	L_2 (μm)	ΔL (μm)	FSR (nm)
1	59.000	27.328	31.672	18.09
2	120.001	27.328	92.673	6.18
3	133.472	27.328	106.144	5.40
4	517.892	27.328	490.564	1.17
5	443.292	27.328	415.964	1.38
6	59.000	27.328	31.672	18.09

The expected FSR values are calculated using $n_g \approx 4.194$ at 1550 nm. All six designs have an FSR below the approximately 50 nm measurement bandwidth. MZI6 repeats MZI1 so that nominally identical devices can be compared after fabrication.

The circuits were simulated in Lumerical INTERCONNECT. Figure 4 shows the two complementary transmission responses for MZI4 around 1550 nm. The periodic fringes arise from the phase difference between the two arms.

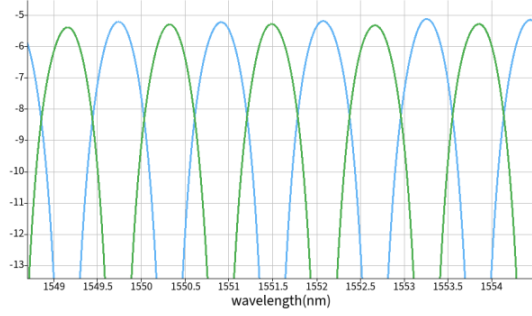


Fig. 4. Simulated MZI4 transmission spectrum around 1550 nm.

For MZI4, $L_1 = 517.892 \mu\text{m}$ and $L_2 = 27.328 \mu\text{m}$, giving $\Delta L = 490.564 \mu\text{m}$. The simulated fringe spacing is approximately 1.18 nm near 1550 nm. Substitution into the FSR equation gives $n_g \approx 4.15$, compared with approximately 4.19 from the MODE waveguide simulation. The difference is about 1.1%, showing that the FSR-based extraction is consistent with the waveguide simulation.

IV. LAYOUT

The six MZIs were implemented in KLayout using the EBeam process design kit. Each circuit includes Y-branch splitters, routed silicon waveguides, and TE grating couplers. The grating couplers are oriented for the course measurement configuration, and the six circuits are placed within the allocated floorplan.

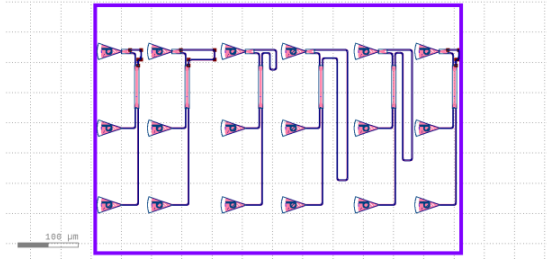


Fig. 5. Proposed KLayout floorplan containing the six MZI circuits.

V. GROUP-INDEX EXTRACTION AND CONCLUSION

In the experiment, the transmission spectrum of each fabricated MZI will be measured through its grating couplers. Adjacent interference maxima or minima will be identified, and their wavelength separation will give the experimental FSR. The known ΔL from the layout will then be substituted into $n_g = \lambda_o^2 / [\Delta L(\text{FSR})]$. The extracted values will be compared with the MODE simulation and with the values obtained from the INTERCONNECT spectra. This provides a direct simulation-to-experiment check of the waveguide group index.

The proposed design therefore provides multiple path-length imbalances while remaining compatible with the available measurement bandwidth. The waveguide model, MZI transfer function, simulated spectrum, and KLayout implementation form the baseline for the later fabrication and experimental analysis stages.

REFERENCES

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